Verification vs Validation

Key Takeaways:

- Donald Firesmith's V-model article has the following diagram that helps place Verifcation & Validation in the context of the SDLC:

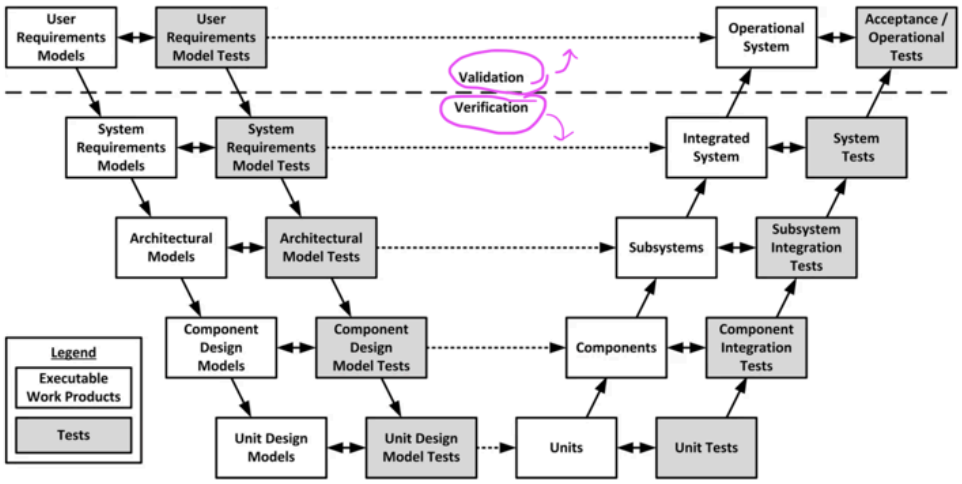


Figure 3: Tester's Double V Model of Testable Work Products and Corresponding Tests

- SDLC timing: Verification activities precede Validation activities because a software product has to be **developed and built** before it can be executed and thus Validated
- Focus: Verification focuses on the software product itself while Validation focuses on the user's experience with the software product
- Reviews count as testing!

Resources

- Verification and validation in software development
  - "**Verification is an ongoing process** that begins long before formal testing and continues until the software application has been reviewed and released. This process involves examining specifications, architecture, and code created during the software development stages."
  - "After verification comes validation."
  - "Validation typically occurs after the entire software development process is complete, assessing whether the client received the product they anticipated. Unlike verification, validation focuses solely on the final product, disregarding internal processes and technical details of the development stages."
  - "**Validation is a one-time process** that begins only after verification is complete."

- System Resilience Part 6: Verification and Validation
  - “Verification is typically performed in one of four ways: Inspection, Analysis, Demonstration, Testing“
  - “Inspection by visual examination includes audits, desk-checking, inspection, walkthroughs, and both formal and informal technical reviews of the system and its relevant technical documentation.“
- Verification Vs Validation - GeeksforGeeks
  - “By clearly defining verification as checking documents and designs without executing code, and validation as testing the actual software for functionality and usability, teams avoid redundant efforts and streamline their testing processes.“
- Verification vs. Validation in Software Testing Processes
  - “Verification testing answers the question: "Are we building the product right?" It checks that the software is being developed in accordance with its specifications.“
  - “Common verification techniques include: Requirements Reviews, Design Reviews, Code Walkthroughs, Unit Testing, Code Inspections (Reviews)“
  - “Validation testing evaluates whether a software product satisfies the needs of its intended users. It answers the critical question: "Are we building the right product?"“
  - “Some common examples of validation testing techniques include: Integration Testing, System Testing“
  - Verification focuses on the software product itself
  - Validation focuses on the user’s experience with the software product
  - “Verification activities can begin as soon as there are artifacts to review, which is often quite early in the project. Requirements documents and design specifications, for example, can be verified as soon as they are written. Validation, on the other hand, typically requires at least a partial working product. It is usually done later in the life cycle, when the software can be tested by real users or in a realistic environment.“
- 5.3 Product Verification
- Fagan Inspection |Professionalqa.com